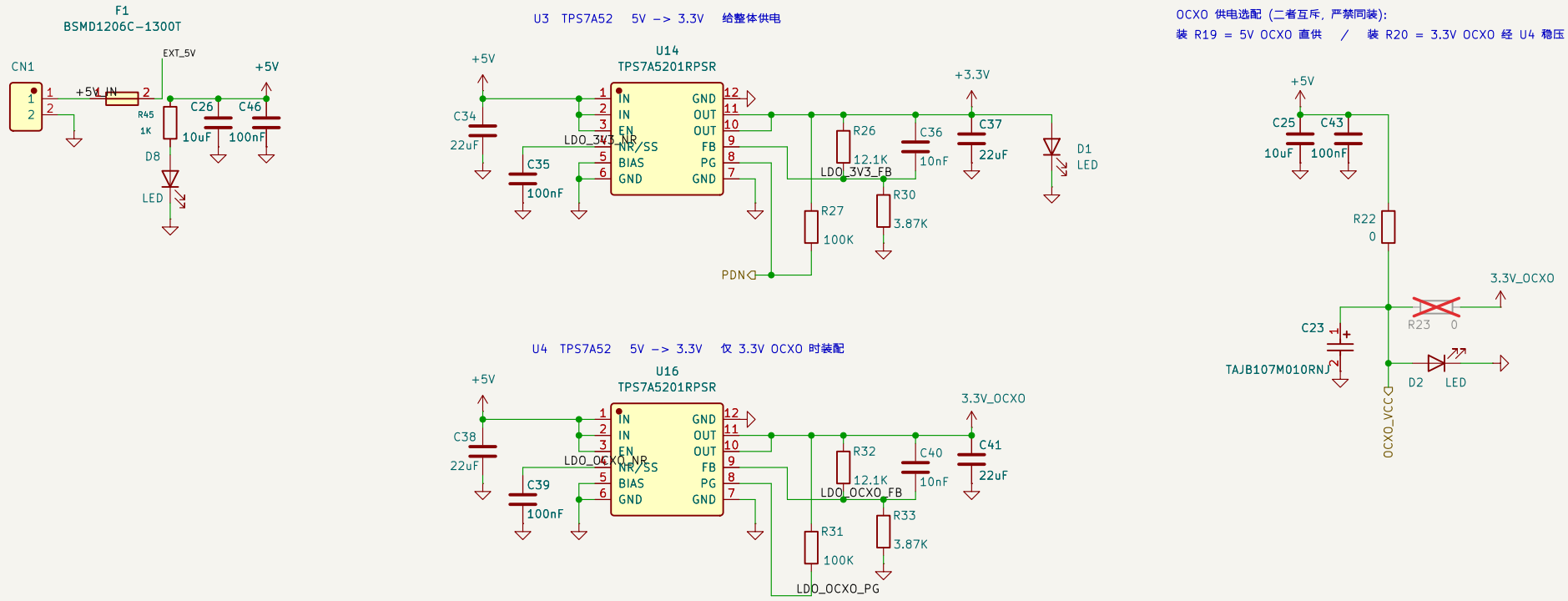

F1
BSMD1206C-1300T
CN1
EXT_5V
+5V_IN
R45
1K
C26
10uF
C46
100nF
+5V
D8
LED
U3 TPS7A52 5V -> 3.3V 给整体供电
U14
TPS7A5201RPSR
+5V
C34
22uF
C35
100nF
LDO_3V3_NR
IN
IN
EN
NR/SS
BIAS
GND
GND
OUT
OUT
FB
PG
GND
R26
12.1K
C36
10nF
+3.3V
C37
22uF
D1
LED
LDO_3V3_FB
R30
3.87K
R27
100K
PDN
OCXO 供电选配 (二者互斥, 严禁同装):
装 R19 = 5V OCXO 直供 / 装 R20 = 3.3V OCXO 经 U4 稳压
+5V
C25
10uF
C43
100nF
R22
0
3.3V_OCXO
R23
0
C23
TAJB107M010RNJ
D2
LED
OCXO_VCC
U4 TPS7A52 5V -> 3.3V 仅 3.3V OCXO 时装配
U16
TPS7A5201RPSR
+5V
C38
22uF
C39
100nF
LDO_OCXO_NR
IN
IN
EN
NR/SS
BIAS
GND
GND
OUT
OUT
FB
PG
GND
R32
12.1K
C40
10nF
3.3V_OCXO
C41
22uF
LDO_OCXO_FB
R33
3.87K
R31
100K
LDO_OCXO_PG
5V source selection: CN1 (priority) / USB VBUS -> +5V
VIN1: EXT_5V after F1. VIN2: USB VBUS. MODE -> VIN1.
PR1 threshold: 4.01V typ. (IC reference +/-8%; not precision UVLO).
-5V generator removed.
TODO: U4 / U12 (BUF602) need a single-supply driver redesign.
PWR_FLAG
EXT_5V
C68
1uF / 16V
GND
VBUS
PWR_FLAG
C69
1uF / 16V
GND
U18
TPS2116DRLR
EXT_5V
MUX_PR1
EXT_5V
VBUS
VIN1
PR1
MODE
VIN2
GND
VOUT
ST
+5V
C70
100nF / 16V
C71
22uF / 10V
GND
EXT_5V
R43
30.1K 1%
MUX_PR1
R44
10K 1%
GND
Inputs: regulated 5V only (max 5.5V). Total mux current <= 2.5A.
USB source must cover OCXO warm-up + full board load.
CC resistors alone do not authorize high-current draw from a PC.
No over-current limit in U18; upstream source protection required.
Check 5V dip on source change under maximum load.
Sheet: /
File: POWER.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1